

Clinical Cloud vs Edge Pipeline Benchmark

Detailed explanation of the patient dataset, cloud pipeline, edge pipeline, transmitted information, security modes, benchmark outputs, and hospital dashboard.

```
Repository: cloud-edge-pipeline-benchmark
Dataset: data/patients.json
Dashboard: http://localhost:8080
Benchmark: docker compose run --rm benchmark
```

1. What This Benchmark Demonstrates

The project compares two ways of processing hospital monitoring data. The cloud pipeline sends the complete patient record to the cloud and performs every clinical processing step there. The edge pipeline processes the same record near the ward or bedside, creates a local early-warning alert, and sends only a reduced clinical summary to the cloud.

Important: the patients are synthetic. They are realistic enough for benchmarking and dashboard design, but they are not real patients and must not be interpreted as medical advice.

2. Patient Dataset

The benchmark uses 20 synthetic patient records. Each record contains demographics, location, diagnosis, comorbidities, medications, and three recent vital-sign samples.

```
patient_id, name, age, sex
ward, bed
primary_diagnosis
comorbidities[]
medications[]
vitals[]:
  timestamp
  heart_rate
  systolic_bp
  respiratory_rate
  spo2
  temperature
  glucose
```

Dataset snapshot

Patient	Ward / Bed	Diagnosis	Latest Vitals
P-0001 - Anna Rossi (78F)	Cardiology / C-12	Heart failure exacerbation	HR 108, SBP 122, RR 25, SpO2 92%, T 37.0, Glu
P-0002 - Marco Bianchi (64M)	Internal Medicine / M-04	Community acquired pneumonia	HR 121, SBP 108, RR 30, SpO2 88%, T 38.7, Glu
P-0003 - Giulia Conti (42F)	Neurology / N-09	Migraine observation	HR 76, SBP 120, RR 16, SpO2 99%, T 36.5, Glu
P-0004 - Luca Ferri (70M)	Emergency / E-02	Suspected sepsis	HR 135, SBP 84, RR 34, SpO2 89%, T 39.5, Glu
P-0005 - Sara Greco (55F)	Surgery / S-18	Post-operative monitoring	HR 95, SBP 121, RR 20, SpO2 96%, T 37.5, Glu
P-0006 - Davide Romano (83M)	Geriatrics / G-07	Dehydration and delirium	HR 115, SBP 96, RR 25, SpO2 94%, T 38.0, Glu
P-0007 - Elena Gallo (35F)	Obstetrics / O-03	Postpartum observation	HR 86, SBP 112, RR 17, SpO2 98%, T 37.0, Glu
P-0008 - Paolo Ricci (59M)	Cardiology / C-04	Acute coronary syndrome observation	HR 107, SBP 140, RR 22, SpO2 95%, T 36.8, Glu
P-0009 - Martina Riva (47F)	Oncology / ON-11	Chemotherapy fever	HR 123, SBP 98, RR 26, SpO2 92%, T 39.1, Glu
P-0010 - Roberto Serra (68M)	Pulmonology / P-15	COPD exacerbation	HR 114, SBP 126, RR 30, SpO2 86%, T 37.8, Glu

Patient	Ward / Bed	Diagnosis	Latest Vitals
P-0011 - Chiara Lombardi (29F)	Emergency / E-08	Syncope observation	HR 72, SBP 110, RR 15, SpO2 99%, T 36.5, Glu
P-0012 - Francesco Moretti (73M)	Nephrology / K-06	Acute kidney injury	HR 101, SBP 94, RR 23, SpO2 95%, T 37.5, Glu
P-0013 - Valentina Neri (51F)	Internal Medicine / M-14	Anemia workup	HR 90, SBP 110, RR 18, SpO2 97%, T 36.8, Glu
P-0014 - Stefano Costa (61M)	ICU Stepdown / I-05	Post-sepsis recovery	HR 109, SBP 106, RR 26, SpO2 92%, T 38.1, Glu
P-0015 - Noemi De Luca (24F)	Surgery / S-03	Appendectomy observation	HR 82, SBP 116, RR 16, SpO2 98%, T 37.0, Glu
P-0016 - Giorgio Fontana (80M)	Stroke Unit / ST-10	Ischemic stroke monitoring	HR 102, SBP 154, RR 22, SpO2 95%, T 36.8, Glu
P-0017 - Irene Marchetti (66F)	Endocrinology / D-02	Hyperglycemia	HR 95, SBP 132, RR 19, SpO2 97%, T 36.8, Glu
P-0018 - Alessio Vitale (57M)	Trauma / T-06	Rib fracture monitoring	HR 108, SBP 118, RR 26, SpO2 92%, T 37.3, Glu
P-0019 - Federica Barbieri (72F)	Cardiology / C-17	Atrial fibrillation rate control	HR 132, SBP 120, RR 23, SpO2 94%, T 37.0, Glu
P-0020 - Matteo Villa (39M)	Infectious Diseases / ID-01	Viral syndrome	HR 96, SBP 116, RR 19, SpO2 97%, T 38.3, Glu

3. Cloud Pipeline in Detail

The cloud pipeline represents a centralized architecture. The complete record is transmitted to the cloud API. This maximizes central visibility but also sends the largest amount of clinical information over the network.

```
benchmark client
-> cloud-api /process
-> simulated network delay
-> schema validation
-> vital-sign normalization and range clamping
-> feature extraction on the full time series
-> vital-sign score
-> trend score
-> clinical context score
-> risk classification and triage recommendation
-> full-record storage
-> response to client
```

Information sent to cloud

- Full demographics: patient identifier, name, age, sex.
- Hospital context: ward and bed.
- Clinical context: diagnosis, comorbidities, medications.
- Complete vital-sign time series: heart rate, systolic blood pressure, respiratory rate, oxygen saturation, temperature, glucose.

Information stored by cloud

- The full patient record.
- The complete vital-sign time series.
- Extracted features: latest values, averages, trends.
- Risk score, risk level, component scores, alert flag, and recommended action.

4. Edge Pipeline in Detail

The edge pipeline represents processing close to the source. The edge receives the same raw record, but it performs early-warning logic locally and sends only the operational result to the cloud. This reduces cloud bandwidth and limits exposure of raw clinical time-series data.

```
benchmark client
-> edge-api /process
-> simulated local network delay
-> schema validation
-> vital-sign range clamping
-> moving-average noise filter
-> local feature extraction
-> early-warning score
-> local alert generation
-> reduced summary sync to cloud-api /sync
-> response to client
```

Information processed locally at the edge

- The complete patient record.
- The complete vital-sign time series.
- Filtered vital-sign samples.
- Local clinical features and risk score.

Information sent from edge to cloud

```
patient_id
```

```
patient_hash pseudonymized from patient_id  
ward  
bed  
latest_vitals  
risk_score  
risk_level  
recommended_action  
alert
```

The edge-to-cloud payload is therefore much smaller than the cloud pipeline payload. It is also less sensitive because it does not include name, medication list, comorbidity list, or the full vital-sign time series.

5. Clinical Processing Logic

The processing logic is implemented in `services/pipeline_api/app/clinical.py`.

Step	Cloud	Edge
Validation	Checks required fields and vital samples.	Same validation before local processing.
Cleaning	Clamps vital signs to plausible ranges.	Clamps vital signs to plausible ranges.
Noise filtering	Uses the raw time series for full central analysis.	Applies moving-average smoothing before scoring.
Feature extraction	Computes latest values, averages, and trends from the full time series.	Computes latest values, averages, and trends from filtered vitals.
Risk scoring	Vital score + trend score + clinical context score.	Same scoring logic on filtered local features.
Alerting	Cloud triage alert after central processing.	Immediate local alert before cloud synchronization.
Storage/sync	Stores full clinical payload.	Syncs reduced summary only.

Risk score components

- Vital score: heart rate, systolic pressure, respiratory rate, SpO2, temperature, glucose.
- Trend score: worsening heart rate, respiratory rate, SpO2, systolic pressure, and temperature across the samples.
- Context score: age, number of comorbidities, and high-risk diagnosis terms such as sepsis, pneumonia, COPD, heart failure, or stroke.
- Risk level: low, medium, high, or critical.

6. Benchmark Scenarios

```
cloud
edge
cloud_simulated_secure
edge_simulated_secure
cloud_tls
edge_tls
```

For every patient, the benchmark runner sends the record to all six scenarios. The simulated secure scenarios add configurable synthetic security costs. The TLS scenarios use real HTTPS/mTLS using local certificates generated by the certgen service.

Scenario	Processing	Transport	Purpose
cloud	Full central processing	HTTP	Baseline centralized pipeline.
edge	Local scoring and reduced sync	HTTP	Baseline edge pipeline.
cloud_simulated_secure	Full central processing	HTTP + simulated security stages	Gain theoretical security overhead.
edge_simulated_secure	Local scoring and reduced sync	HTTP + simulated security stages	Gain security overhead on edge and sync.
cloud_tls	Full central processing	HTTPS/mTLS	Measure real transport security overhead.
edge_tls	Local scoring and reduced sync	HTTPS/mTLS	Measure TLS on client-edge and edge-cloud sync.

7. Benchmark Outputs

```
results/results.csv
results/summary.json
results/patient_results.json
```

`results.csv` is the technical benchmark table. `summary.json` aggregates latency, payload, and alert counts by scenario. `patient_results.json` is optimized for the hospital dashboard.

Field	Meaning
patient_id, ward, bed, diagnosis	Clinical context of the processed patient.
client_total_ms	End-to-end latency seen by the benchmark client.
service_total_ms	Time measured inside the API handler.
client_service_delta_ms	Client-side overhead not measured inside the service. TLS handshake and HTTP overhead mainly appear here.
network_ms	Configured network delay stage.
preprocess_ms	Validation, cleaning, filtering, and feature preparation stage.
inference_ms	Clinical risk scoring stage.
sync_ms	Edge-to-cloud synchronization time.
payload_sent_kb	Input payload sent by the benchmark client.
payload_synced_kb	Payload sent to cloud. Full record in cloud mode, reduced summary in edge mode.
risk_score, risk_level	Computed patient risk.
recommended_action	Operational recommendation for the ward team.

8. Hospital Dashboard

The hospital dashboard is served by `services/hospital_dashboard/app/main.py` at `http://localhost:8080`.

- Patient worklist sorted by risk.
- Ward and risk filters.
- Latest vital signs for the selected patient.
- Diagnosis, medications, comorbidities, risk score, and recommended action.
- Per-patient comparison of cloud, edge, simulated security, and TLS scenarios.
- Ward load and active alert count.

The dashboard reads `patient_results.json` after the benchmark runs. If the benchmark has not been executed yet, it still displays the dataset and marks patients as pending.

9. Prometheus and Grafana

Prometheus collects runtime metrics from the API services. It does not execute the benchmark. Grafana visualizes those metrics over time. The hospital dashboard is separate: it visualizes clinical benchmark results.

```
pipeline_requests_total
pipeline_total_latency_ms
pipeline_stage_latency_ms
pipeline_payload_size_kb
pipeline_in_flight_requests
```

10. Commands

```
docker compose up --build
docker compose run --rm benchmark
docker compose ps
docker compose logs -f cloud-api edge-api cloud-api-tls edge-api-tls hospital-dashboard
docker compose down
```

11. Limitations

- The dataset is synthetic and small by design.
- The scoring model is deterministic and explanatory; it is not a validated medical model.
- The edge is emulated through Docker CPU/RAM limits, not real bedside hardware.
- TLS certificates are local self-signed certificates for benchmarking, not production certificates.
- Network delay is simulated at application level; a stronger network study could use tc netem, Mininet, or Containernet.